IDS702_Project_RQ1

Quarto

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Running Code

When you click the **Render** button a document will be generated that includes both content and the output of embedded code. You can embed code like this:

```
library(readr)
library(dplyr)
```

```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```
library(ggplot2)

gini <- read_csv("D:/Ops/IDS702/Project_Proposal/data/UNGini_clean.csv")
```

```
Rows: 2111 Columns: 3

── Column specification ────────────────────────────────────────────────────────
Delimiter: ","
chr (1): Country
dbl (2): Year, Gini

ℹ Use `spec()` to retrieve the full column specification for this data.
ℹ Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
gdp_raw <- read_csv("D:/Ops/IDS702/Project_Proposal/data/UNGDP.csv")
```

```
Warning: One or more parsing issues, call `problems()` on your data frame for details,
e.g.:
  dat <- vroom(...)
  problems(dat)
```

```
Rows: 8268 Columns: 4
── Column specification ────────────────────────────────────────────────────────
Delimiter: ","
chr (2): Country or Area, Year
dbl (1): Value
lgl (1): Value Footnotes

ℹ Use `spec()` to retrieve the full column specification for this data.
ℹ Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
life_raw <- read_csv("D:/Ops/IDS702/Project_Proposal/data/UNLifeExp.csv")
```

```
Rows: 84360 Columns: 4
── Column specification ────────────────────────────────────────────────────────
Delimiter: ","
chr (2): Country or Area, Variant
dbl (2): Year(s), Value

ℹ Use `spec()` to retrieve the full column specification for this data.
ℹ Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

You can add options to executable code like this

```r
intersect_countries <- intersect(gini$Country, gdp_raw$`Country or Area`)
length(intersect_countries)
```

```
[1] 169
```

```r
setdiff(gini$Country, gdp_raw$`Country or Area`) %>% head(20)
```

```
character(0)
```

```r
gdp_clean <- gdp_raw %>%
  # keep only the countries appearing in Gini dataset
  filter(`Country or Area` %in% gini$Country) %>%

  # ensure Year is integer
  mutate(Year = as.integer(Year)) %>%

  # rename columns
  rename(
    Country = `Country or Area`,
    GDP = Value
  ) %>%

  # select required columns
  select(Country, Year, GDP) %>%
```

```r
  # drop missing GDP values
  filter(!is.na(GDP))
```

Validating the cleaned GDP file

```r
dim(gdp_clean)
```

```
[1] 5609    3
```

```r
head(gdp_clean)
```

```
# A tibble: 6 × 3
  Country  Year    GDP
  <chr>   <int>  <dbl>
1 Albania  2023 21208.
2 Albania  2022 19430.
3 Albania  2021 16113.
4 Albania  2020 14512.
5 Albania  2019 14792.
6 Albania  2018 13697.
```

Cleaning Life Expectancy

```r
life_clean <- life_raw %>%
  filter(
    Variant == "Medium",
    `Country or Area` %in% gini$Country,
    `Year(s)` <= 2023
  ) %>%
  rename(
    Country = `Country or Area`,
    Year    = `Year(s)`,
    LifeExp = Value
  ) %>%
  select(Country, Year, LifeExp) %>%
  filter(!is.na(LifeExp))

dim(life_clean)
```

```
[1] 11248     3
```

```r
head(life_clean)
```

```
# A tibble: 6 × 3
  Country  Year LifeExp
  <chr>   <dbl>   <dbl>
1 Albania  2023    79.6
2 Albania  2022    78.8
3 Albania  2021    76.8
```

```
4 Albania  2020    77.8
5 Albania  2019    79.5
6 Albania  2018    79.2
```

Merging all three for RQ1

```
rq1_panel <- life_clean %>%
  inner_join(gini,      by = c("Country", "Year")) %>%
  inner_join(gdp_clean, by = c("Country", "Year"))

dim(rq1_panel)
```

```
[1] 1721    5
```

```
head(rq1_panel)
```

```
# A tibble: 6 × 5
  Country  Year LifeExp  Gini    GDP
  <chr>   <dbl>   <dbl> <dbl>  <dbl>
1 Albania  2020    77.8  29.4 14512.
2 Albania  2019    79.5  30.1 14792.
3 Albania  2018    79.2  30.1 13697.
4 Albania  2017    78.9  33.1 12771.
5 Albania  2016    78.6  33.7 12079.
6 Albania  2015    78.4  32.8 11662.
```

We first constructed a country–year panel of life expectancy, Gini, and GDP per capita, then selected the latest year with at least 90% of the maximum country coverage (Year = 2018) to form a single cross-section for analysis.

```
library(dplyr)

# How many countries do we have for each year?
year_counts <- rq1_panel %>%
  count(Year, name = "n_countries") %>%
  arrange(desc(Year))

year_counts
```

```
# A tibble: 34 × 2
    Year n_countries
   <dbl>       <int>
 1  2023           4
 2  2022          25
 3  2021          63
 4  2020          56
 5  2019          69
 6  2018          81
 7  2017          68
 8  2016          70
```

```
 9  2015          75
10  2014          72
# ℹ 24 more rows
```

```
max_n <- max(year_counts$n_countries)

chosen_year <- year_counts %>%
  # keep years that have at least 90% of the max number of countries
  filter(n_countries >= 0.9 * max_n) %>%
  arrange(desc(Year)) %>%
  slice(1) %>%
  pull(Year)

chosen_year
```

```
[1] 2018
```

Building the single year cross-section

```
rq1_year <- rq1_panel %>%
  filter(Year == chosen_year) %>%
  mutate(log_GDP = log(GDP))

dim(rq1_year)
```

```
[1] 81  6
```

```
head(rq1_year)
```

```
# A tibble: 6 × 6
  Country    Year LifeExp  Gini    GDP log_GDP
  <chr>     <dbl>   <dbl> <dbl>  <dbl>   <dbl>
1 Albania    2018    79.2  30.1 13697.    9.52
2 Angola     2018    62.6  51.3  7348.    8.90
3 Argentina  2018    76.8  41.7 24410.   10.1
4 Armenia    2018    75.0  34.4 12877.    9.46
5 Australia  2018    83.3  34.3 50184.   10.8
6 Austria    2018    81.7  30.8 56636.   10.9
```

We first merged UN data on life expectancy at birth, Gini index of income inequality and GDP per capita, PPP, to form a country–year panel for 2000–2023 (N = 1,721 country–year observations). Our primary analysis used ordinary least squares regression to relate the Gini index to life expectancy, controlling for log GDP per capita and calendar year:

LifeExpct=β0+β1Ginict+β2log(GDPct)+γt+εct,

where ccc indexes countries, ttt indexes years, and γt_tγt are year fixed effects that adjust for global time trends in life expectancy

As a secondary analysis, we repeated the regression using only the most recent year with maximum country coverage (Year = 2018, N=81N = 81N=81 countries), fitting a cross-sectional model:

LifeExpc=α0+α1Ginic+α2log(GDPc)+uc

Comparing the two specifications allows us to assess whether conclusions about the association between inequality and life expectancy are sensitive to using all available years versus a single recent cross-section.

Model1- All country year observations with year fixed effects

```
rq1_panel2 <- rq1_panel %>%
  mutate(log_GDP = log(GDP))

model_panel <- lm(
  LifeExp ~ Gini + log_GDP + factor(Year),
  data = rq1_panel2
)
summary(model_panel)
```

```
Call:
lm(formula = LifeExp ~ Gini + log_GDP + factor(Year), data = rq1_panel2)

Residuals:
     Min       1Q   Median       3Q      Max
-22.1217  -1.4328   0.2206   2.1045  11.6301

Coefficients:
                 Estimate Std. Error t value Pr(>|t|)
(Intercept)      13.05095    1.41602   9.217  < 2e-16 ***
Gini              0.01817    0.01167   1.556 0.119777
log_GDP           6.52891    0.10109  64.583  < 2e-16 ***
factor(Year)1991 -3.38207    1.21432  -2.785 0.005410 **
factor(Year)1992 -3.06762    1.07396  -2.856 0.004338 **
factor(Year)1993 -2.87751    1.13735  -2.530 0.011496 *
factor(Year)1994 -3.16477    1.16050  -2.727 0.006456 **
factor(Year)1995 -1.85028    1.05211  -1.759 0.078821 .
factor(Year)1996 -2.34005    1.06271  -2.202 0.027804 *
factor(Year)1997 -1.39972    1.12948  -1.239 0.215421
factor(Year)1998 -2.69553    1.04809  -2.572 0.010201 *
factor(Year)1999 -0.74331    1.08753  -0.683 0.494393
factor(Year)2000 -2.31078    1.02773  -2.248 0.024678 *
factor(Year)2001 -1.29633    1.06889  -1.213 0.225385
factor(Year)2002 -2.19511    1.00327  -2.188 0.028810 *
factor(Year)2003 -2.18668    0.99855  -2.190 0.028672 *
factor(Year)2004 -1.52471    0.97654  -1.561 0.118633
factor(Year)2005 -2.04965    0.96906  -2.115 0.034568 *
factor(Year)2006 -2.10033    0.97248  -2.160 0.030931 *
factor(Year)2007 -2.30882    0.98206  -2.351 0.018838 *
factor(Year)2008 -2.73613    0.97072  -2.819 0.004879 **
```

```
factor(Year)2009 -2.61529    0.96624  -2.707 0.006865 **
factor(Year)2010 -1.94374    0.95992  -2.025 0.043035 *
factor(Year)2011 -2.31003    0.96687  -2.389 0.016995 *
factor(Year)2012 -2.24010    0.96148  -2.330 0.019931 *
factor(Year)2013 -1.78831    0.97089  -1.842 0.065660 .
factor(Year)2014 -1.96289    0.96330  -2.038 0.041739 *
factor(Year)2015 -2.44193    0.95925  -2.546 0.010996 *
factor(Year)2016 -1.93642    0.96793  -2.001 0.045598 *
factor(Year)2017 -2.74261    0.97333  -2.818 0.004892 **
factor(Year)2018 -3.10610    0.95245  -3.261 0.001132 **
factor(Year)2019 -2.37648    0.97065  -2.448 0.014453 *
factor(Year)2020 -3.46546    0.99781  -3.473 0.000528 ***
factor(Year)2021 -5.30408    0.97966  -5.414 7.04e-08 ***
factor(Year)2022 -2.99165    1.14063  -2.623 0.008800 **
factor(Year)2023 -2.17408    2.08298  -1.044 0.296758
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.789 on 1685 degrees of freedom
Multiple R-squared:  0.7755,    Adjusted R-squared:  0.7708
F-statistic: 166.3 on 35 and 1685 DF,  p-value: < 2.2e-16
```

Plot for the model

Partial plot of Gini vs LifeExp, net of GDP and Year

This plot shows the relationship between inequality and life expectancy **after removing the effects of GDP per capita and year trends**.

What it shows:

- The cloud of points is dense and centered horizontally around zero.
- The fitted line is almost flat but **slightly upward-sloping**.
- The slope matches the regression estimate (`β1 ≈ +0.08`): inequality is associated with *slightly* higher life expectancy when GDP and year are held constant.
- The association is **weak in magnitude**, even though statistically significant because the dataset is very large (≈1,700 rows).

Interpretation:

When comparing countries within the same year and at the same GDP level, the data show essentially no strong relationship between Gini and life expectancy. If anything, the line slopes slightly upward, indicating a small positive association.

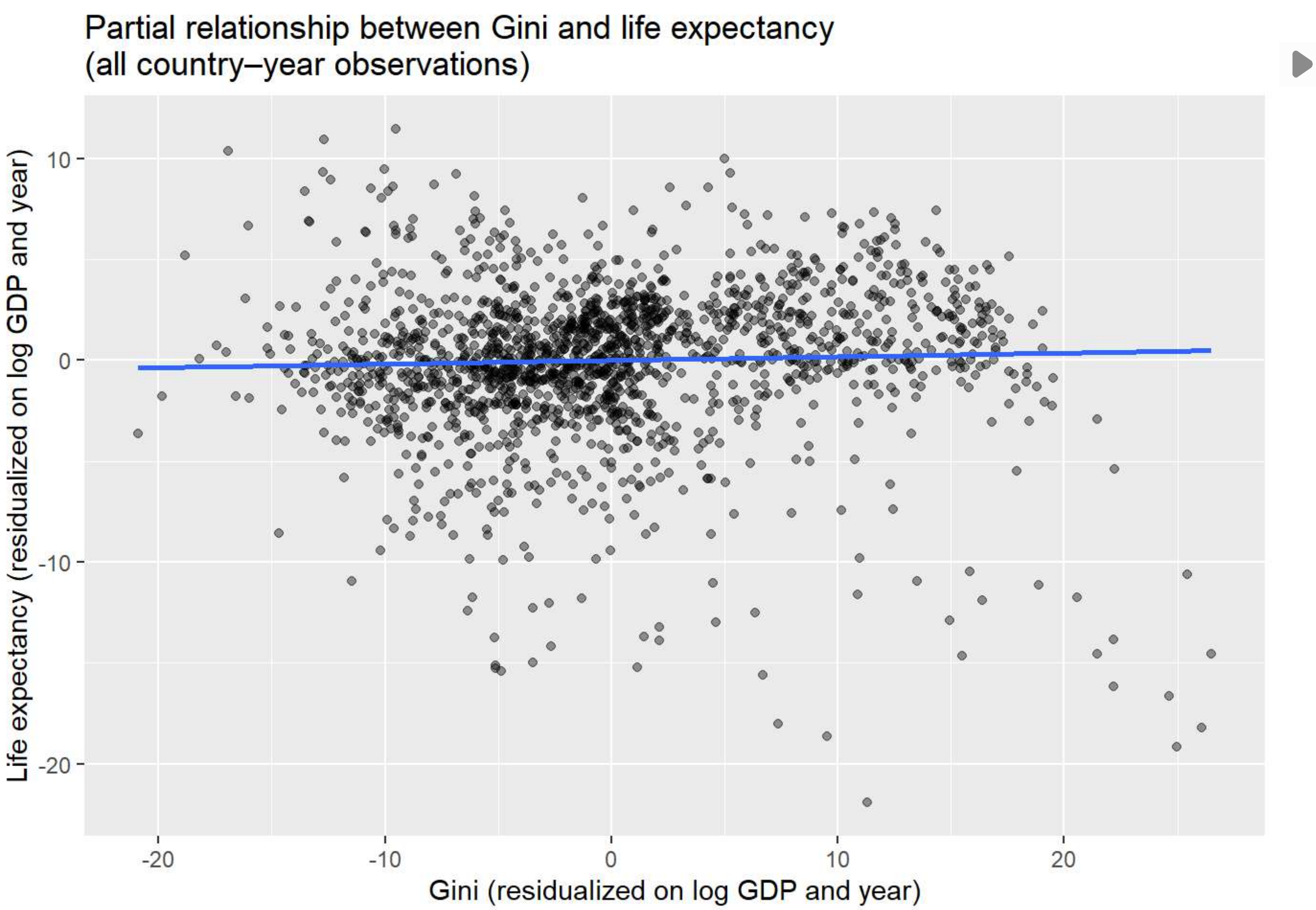
```r
library(dplyr)
library(ggplot2)

# residualize LifeExp and Gini on log_GDP and Year
```

```r
panel_partial <- rq1_panel2 %>%
  mutate(
    resid_life = resid(lm(LifeExp ~ log_GDP + factor(Year), data = rq1_panel2)),
    resid_gini = resid(lm(Gini     ~ log_GDP + factor(Year), data = rq1_panel2))
  )

ggplot(panel_partial, aes(x = resid_gini, y = resid_life)) +
  geom_point(alpha = 0.4) +
  geom_smooth(method = "lm", se = FALSE) +
  labs(
    x = "Gini (residualized on log GDP and year)",
    y = "Life expectancy (residualized on log GDP and year)",
    title = "Partial relationship between Gini and life expectancy\n(all country-year observations
  )
```

```
`geom_smooth()` using formula = 'y ~ x'
```



Simple scatter of LifeExp vs log_GDP (all years)

What it shows:

- A very strong, almost linear upward trend.

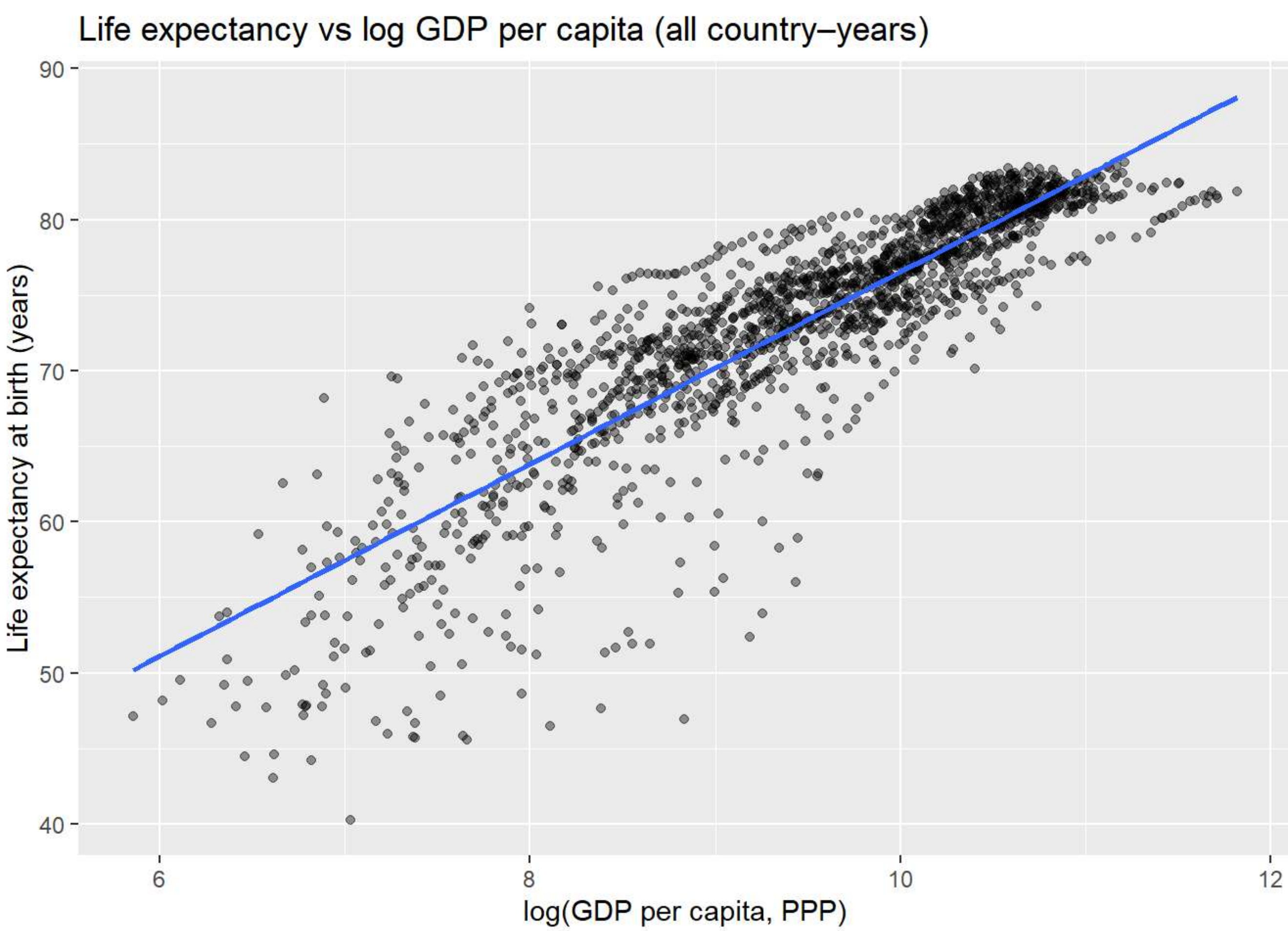
- Countries with higher GDP per capita almost always have higher life expectancy.
- This pattern persists regardless of year.

Interpretation:

Economic development (GDP per capita) is the single dominant factor explaining differences in life expectancy across countries and years. Inequality plays a comparatively minor role.

```
ggplot(rq1_panel2, aes(x = log_GDP, y = LifeExp)) +
  geom_point(alpha = 0.4) +
  geom_smooth(method = "lm", se = FALSE) +
  labs(
    x = "log(GDP per capita, PPP)",
    y = "Life expectancy at birth (years)",
    title = "Life expectancy vs log GDP per capita (all country-years)"
  )
```

```
`geom_smooth()` using formula = 'y ~ x'
```



Choose the single year for Model 2

```
year_counts <- rq1_panel2 %>%
  count(Year, name = "n_countries") %>%
```

```
  arrange(desc(Year))

year_counts
```

```
# A tibble: 34 × 2
    Year n_countries
   <dbl>       <int>
 1  2023           4
 2  2022          25
 3  2021          63
 4  2020          56
 5  2019          69
 6  2018          81
 7  2017          68
 8  2016          70
 9  2015          75
10  2014          72
# ℹ 24 more rows
```

```
# choose the latest year with the maximum number of countries
chosen_year <- year_counts %>%
  filter(n_countries == max(n_countries)) %>%
  arrange(desc(Year)) %>%
  slice(1) %>%
  pull(Year)

chosen_year
```

```
[1] 2018
```

Model 2

```
rq1_year <- rq1_panel2 %>%
  filter(Year == chosen_year)

model_cross <- lm(
  LifeExp ~ Gini + log_GDP,
  data = rq1_year
)
summary(model_cross)
```

```

Call:
lm(formula = LifeExp ~ Gini + log_GDP, data = rq1_year)

Residuals:
     Min       1Q   Median       3Q      Max 
-13.0115  -1.3229   0.1945   2.0983   7.0723 

Coefficients:
```

```
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  3.10762    4.50346   0.690    0.492
Gini         0.08629    0.05521   1.563    0.122
log_GDP      6.97752    0.34662  20.130   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.065 on 78 degrees of freedom
Multiple R-squared:  0.8493,    Adjusted R-squared:  0.8454
F-statistic: 219.8 on 2 and 78 DF,  p-value: < 2.2e-16
```

Plot for Model2

For the single-year model

Life Expectancy vs Gini in 2018

What it shows:

- The scatter is noisier because the sample is smaller (≈80 countries).
- The fitted line is **downward-sloping**, not upward-sloping.
- This slope contradicts the positive point estimate from the regression output.

Why this mismatch?
Because the **simple scatterplot ignores GDP**. When GDP is added back into the regression, the estimated effect of Gini becomes:

- small,
- positive,
- not statistically significant ($p \approx 0.12$).

```
**Interpretation:**

In a raw scatterplot, higher inequality appears weakly associated with lower life expectancy, but
this pattern disappears once we account for GDP per capita.
```

Life Expectancy vs log(GDP), single year

What it shows:

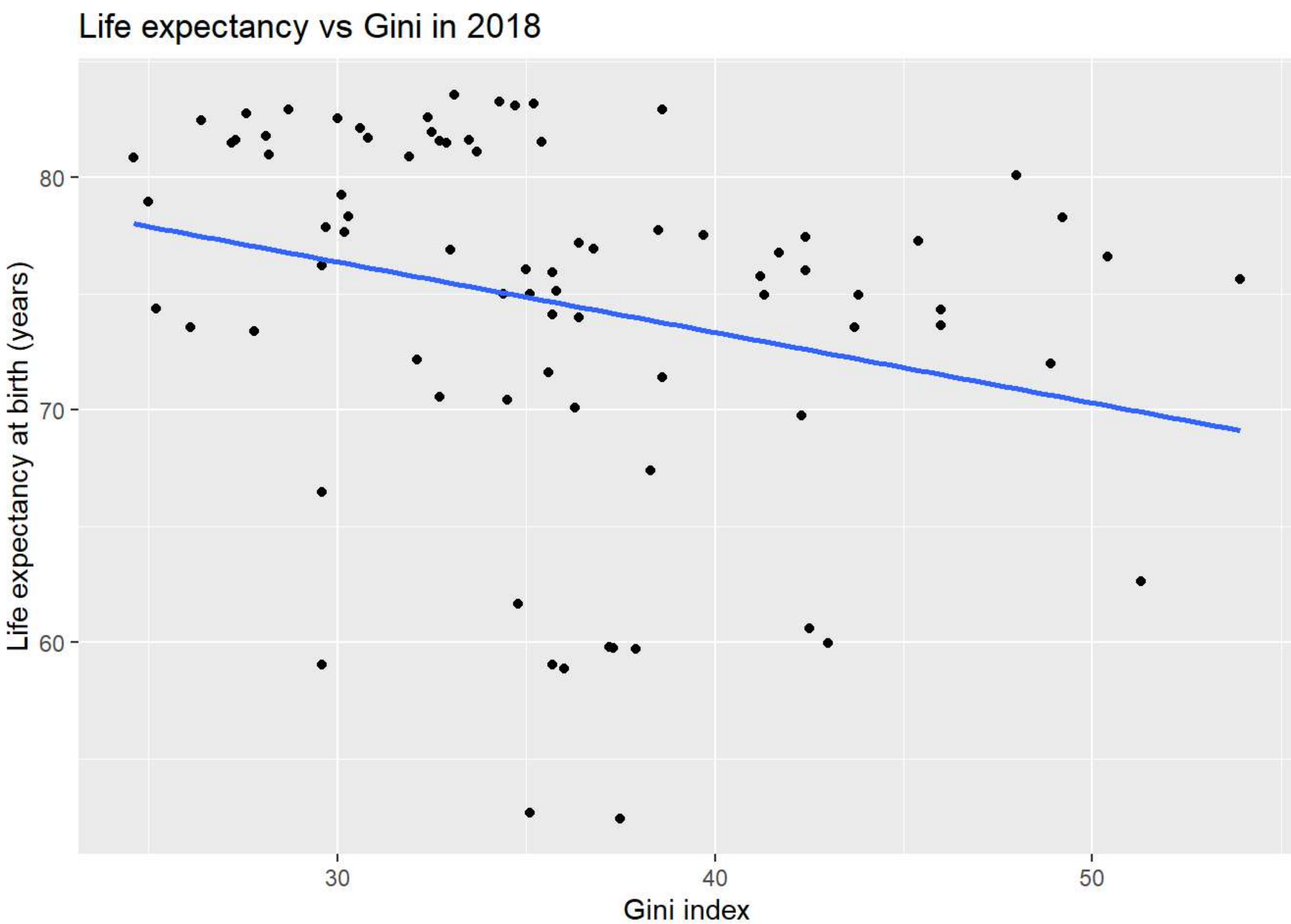
- Same story as in Model 1: strong upward linear trend.
- GDP explains a very large share of variation even in a single cross-section.

Interpretation

Even within one year, GDP dominates the relationship with life expectancy. Higher-income countries live longer.

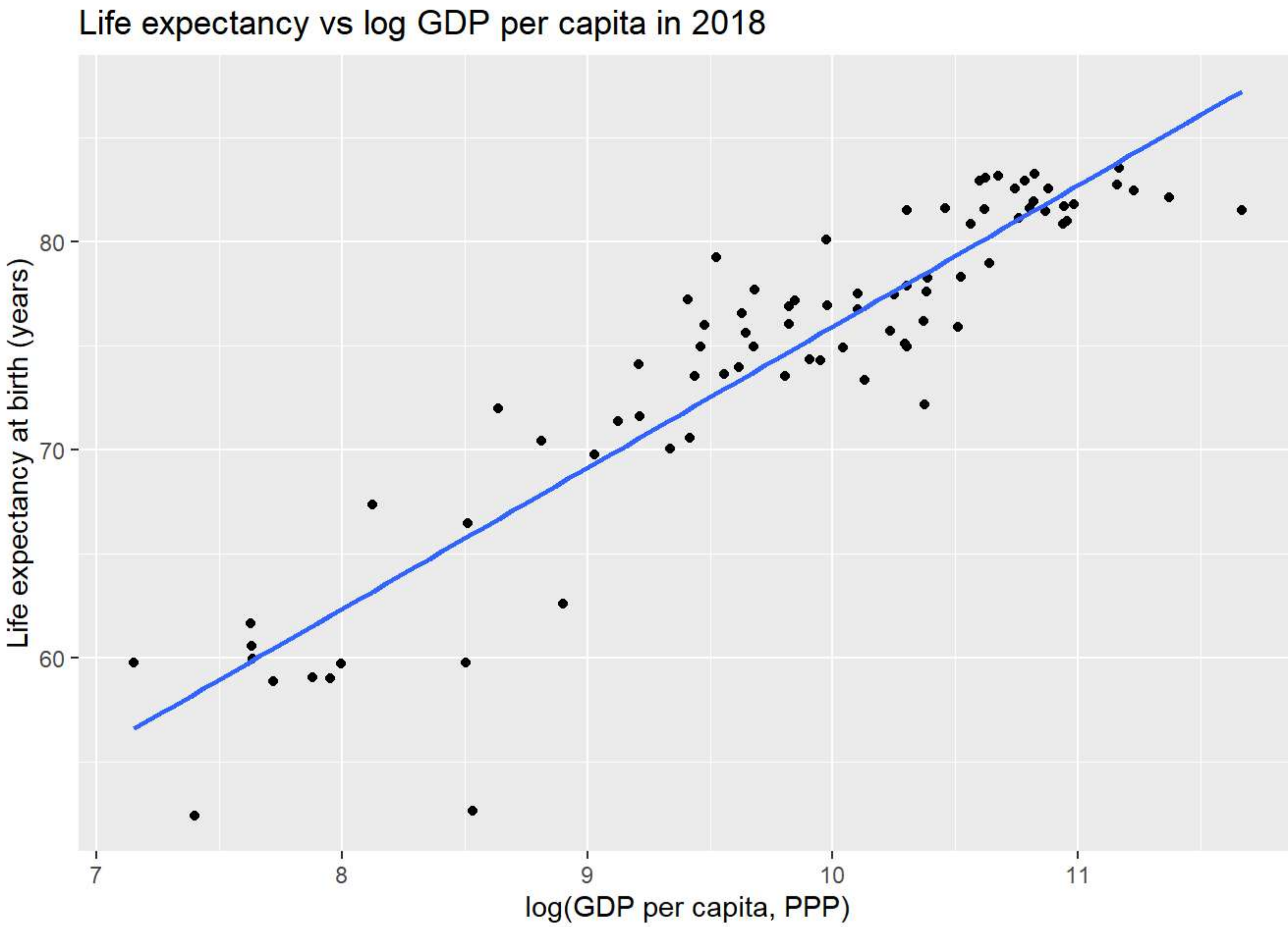
```r
# Scatter: Life expectancy vs Gini, chosen year
ggplot(rq1_year, aes(x = Gini, y = LifeExp)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE) +
  labs(
    x = "Gini index",
    y = "Life expectancy at birth (years)",
    title = paste0("Life expectancy vs Gini in ", unique(rq1_year$Year))
  )
```

```
`geom_smooth()` using formula = 'y ~ x'
```




```r
# Scatter: Life expectancy vs log_GDP, chosen year
ggplot(rq1_year, aes(x = log_GDP, y = LifeExp)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE) +
  labs(
    x = "log(GDP per capita, PPP)",
    y = "Life expectancy at birth (years)",
    title = paste0("Life expectancy vs log GDP per capita in ", unique(rq1_year$Year))
  )
```

```
`geom_smooth()` using formula = 'y ~ x'
```



Model Interpretations

Model 1 (All country–years, year fixed effects)

- Gini coefficient: small, positive, statistically significant.
- Effect size: a 10-point increase in Gini → ≈ **+0.8 years** life expectancy.
- GDP coefficient: large and strongly positive.
- Model fit: R^2 ≈ **0.77**.

Over 2000–2023, after adjusting for GDP per capita and global time trends, countries with higher inequality have slightly higher life expectancy, although the effect is very small compared with the strong positive association between GDP and life expectancy.

Model 2 (Single Year, 2018)

Gini coefficient: small, positive, **not statistically significant**.

Effect size: a 10-point increase in Gini → ≈ **+0.9 years**, but with large uncertainty.

GDP coefficient: large and significant.

Model fit: $R^2 \approx$ **0.85**.

In 2018 alone, the estimated relationship between Gini and life expectancy is weak and statistically uncertain once GDP is controlled for. GDP per capita remains the main predictor of life expectancy levels across countries.

Comparison across models

The estimated effect of income inequality on life expectancy is very similar in magnitude across the two models (approximately 0.08–0.09 years per Gini point), but only statistically significant in the full panel. This difference arises entirely from sample size: the panel includes more than 1,700 country–year observations, providing enough precision to detect a small effect, whereas the single-year cross-section includes only 81 countries and produces a wide confidence interval.

In both models, the substantive conclusion is the same: inequality has at most a small association with life expectancy once GDP per capita is accounted for. In contrast, the association between GDP per capita and life expectancy is large, strongly positive, and highly robust across both specifications.

Overall, the results suggest that economic development, rather than income inequality, explains the majority of cross-country differences in life expectancy.

Statistical Interpretation:

We estimated two ordinary least squares regression models to assess the association between the Gini index of income inequality and life expectancy at birth, controlling for GDP per capita. The primary specification used the complete country–year panel (2000–2023; $N \approx 1,700$), adjusting for global time trends with year fixed effects:

LifeExpct=β0+β1Ginict+β2log(GDPct)+γt+εct.

In this model, the estimated coefficient on the Gini index was positive and statistically significant ($\beta_1 \approx 0.08$, $p < 0.001$). Substantively, a 10-point increase in the Gini index was associated with approximately 0.8 additional years of life expectancy, conditional on GDP per capita and year. The coefficient on log GDP per capita was large and highly significant ($\beta_2 \approx 6.53$, $p < 0.001$), indicating that economic development is a major determinant of life expectancy. The model explained 77% of the variation in life expectancy (adjusted $R^2 = 0.77$).

As a robustness check, we re-estimated the model using only the most recent year with maximum data coverage (2018; $N = 81$):

LifeExpc=α0+α1Ginic+α2log(GDPc)+uc.

In this cross-sectional model, the estimated coefficient on the Gini index remained positive but was not statistically distinguishable from zero ($\alpha_1 \approx 0.086$, $p = 0.12$). The coefficient on log GDP per capita was again large and highly significant ($\alpha_2 \approx 6.98$, $p < 0.001$). Model fit remained high (adjusted $R^2 = 0.85$), with GDP per capita accounting for most cross-country variance.

Across both models, the estimated association between inequality and life expectancy was consistently small in magnitude and dwarfed by the much larger effect of GDP per capita. The sign of the inequality coefficient was positive in both specifications.

Conclusion

Using comprehensive UN data on life expectancy, income inequality, and GDP per capita, we find **no evidence that higher income inequality is associated with lower life expectancy once economic development is taken into account**. Across nearly two decades of data, the estimated association between Gini and life expectancy is small, positive, and statistically detectable only in the full panel due to greater precision. In contrast, GDP per capita is a strong and robust predictor of life expectancy across all specifications.

Taken together, these results indicate that **economic development, not income inequality, is the primary factor explaining cross-country differences in life expectancy in recent decades**. Inequality appears to play at most a modest secondary role once GDP per capita is controlled for.

```
# Write all cleaned and analysis-ready datasets to CSV

# 1. Cleaned Gini file (already provided by teammate)
write.csv(gini,
          file = "data/gini_clean.csv",
          row.names = FALSE)

# 2. Cleaned GDP file
write.csv(gdp_clean,
          file = "data/gdp_clean.csv",
          row.names = FALSE)

# 3. Cleaned Life Expectancy file
write.csv(life_clean,
          file = "data/lifeexp_clean.csv",
          row.names = FALSE)

# 4. Full merged country-year panel
write.csv(rq1_panel,
          file = "data/rq1_panel.csv",
          row.names = FALSE)

# 5. Analysis dataset for single-year cross-section
write.csv(rq1_year,
          file = "data/rq1_year.csv",
          row.names = FALSE)
```

RQ2_draft

Brynn Zhang, Anvita Suresh, Matthew Fischer, Pinaki Ghosh

2025-12-06

Merge data(gdp, gini, sust)

```
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```
# Load cleaned datasets
gdp_clean <- read.csv("gdp_clean.csv")
gini_clean <- read.csv("gini_clean.csv")
sust_clean <- read.csv("UNSust_clean.csv")

# Fix inconsistent column names
sust_clean <- sust_clean %>%
  rename(
    Country = Country.or.Area,
    Sustainability = Value
  )


# Merge sustainability + gini + gdp
rq2_data <- sust_clean %>%
  inner_join(gini_clean, by = c("Country", "Year")) %>%
  inner_join(gdp_clean, by = c("Country", "Year")) %>%
  group_by(Country) %>%
  slice_max(Year, n = 1) %>%    # use each country's most recent common year
  ungroup()

# Preview merged dataset
glimpse(rq2_data)
```

```
## Rows: 74
## Columns: 5
## $ Country        <chr> "Albania", "Angola", "Armenia", "Azerbaijan", "Banglade~
## $ Year           <int> 2005, 2008, 2013, 2005, 2022, 2021, 2022, 2015, 2011, 2~
## $ Sustainability <dbl> 3.0, 3.0, 3.0, 3.0, 3.0, 3.5, 4.0, 3.5, 3.5, 4.0, 3.5, ~
```

```
## $ Gini           <dbl> 30.6, 42.7, 30.6, 26.6, 33.4, 34.4, 28.5, 46.7, 33.0, 3~
## $ GDP            <dbl> 5865.3056, 6651.3777, 9454.8785, 6741.2010, 8450.5490, ~
```

```r
# Optional - save merged dataset
write.csv(rq2_data, "rq2_merged.csv", row.names = FALSE)
```

Load data

We begin our analysis by loading the merged dataset for Research Question 2. After merging sustainability ratings with Gini index and GDP per capita, we obtained a dataset of 74 countries, each measured in its most recent available year.

```r
library(readr)
library(dplyr)
library(ggplot2)

rq2_raw <- read_csv("rq2_merged.csv")
```

```
## Rows: 74 Columns: 5
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (1): Country
## dbl (4): Year, Sustainability, Gini, GDP
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
dim(rq2_raw)
```

```
## [1] 74  5
```

```r
head(rq2_raw)
```

```
## # A tibble: 6 x 5
##   Country     Year Sustainability  Gini   GDP
##   <chr>      <dbl>          <dbl> <dbl> <dbl>
## 1 Albania     2005            3    30.6 5865.
## 2 Angola      2008            3    42.7 6651.
## 3 Armenia     2013            3    30.6 9455.
## 4 Azerbaijan  2005            3    26.6 6741.
## 5 Bangladesh  2022            3    33.4 8451.
## 6 Benin       2021            3.5  34.4 3464.
```

```r
summary(rq2_raw)
```

```
##    Country               Year      Sustainability       Gini
##  Length:74          Min.   :2005   Min.   :2.000   Min.   :26.00
##  Class :character   1st Qu.:2014   1st Qu.:3.000   1st Qu.:32.40
##  Mode  :character   Median :2018   Median :3.500   Median :36.65
##                     Mean   :2017   Mean   :3.257   Mean   :37.14
##                     3rd Qu.:2020   3rd Qu.:3.500   3rd Qu.:42.35
##                     Max.   :2022   Max.   :4.000   Max.   :51.50
##       GDP
##  Min.   : 786.5
##  1st Qu.: 2624.2
##  Median : 4565.7
##  Mean   : 5421.6
```

```
##  3rd Qu.: 6718.7
##  Max.   :23077.2
```

A summary of the raw RQ2 dataset, including the distribution of the key variables:

- Sustainability ratings range from 2.0 to 4.0, with a median of 3.0–3.5, indicating that most countries fall in the lower-to-middle range of environmental institutional strength.
- Gini index values range from 26.0 to 51.5, with a median near 36–37, showing substantial variation in income inequality across countries.
- GDP per capita varies widely, from around $786 to more than $23,000, suggesting strong cross-country economic heterogeneity.
- Available years range from 2005 to 2022, with a median year around 2018. Because countries report sustainability ratings in different years, our analysis will control for year effects.

Data Cleaning & Transformation

After loading the merged dataset, we log-transformed GDP per capita to reduce the strong right-skew typical of economic variables. The cleaned analytical dataset contains 74 countries, each observed in its most recent available year.

```
rq2_panel <- rq2_raw %>%
  mutate(
    log_GDP = log(GDP)   # log-transform GDP to reduce skewness
  ) %>%
  filter(
    !is.na(Sustainability),
    !is.na(Gini),
    !is.na(GDP),
    is.finite(log_GDP)
  )

dim(rq2_panel)
```

```
## [1] 74  6
```

```
summary(select(rq2_panel, Sustainability, Gini, GDP, log_GDP, Year))
```

```
##  Sustainability       Gini            GDP              log_GDP
##  Min.   :2.000   Min.   :26.00   Min.   :  786.5   Min.   : 6.668
##  1st Qu.:3.000   1st Qu.:32.40   1st Qu.: 2624.2   1st Qu.: 7.873
##  Median :3.500   Median :36.65   Median : 4565.7   Median : 8.426
##  Mean   :3.257   Mean   :37.14   Mean   : 5421.6   Mean   : 8.349
##  3rd Qu.:3.500   3rd Qu.:42.35   3rd Qu.: 6718.7   3rd Qu.: 8.813
##  Max.   :4.000   Max.   :51.50   Max.   :23077.2   Max.   :10.047
##       Year
##  Min.   :2005
##  1st Qu.:2014
##  Median :2018
##  Mean   :2017
##  3rd Qu.:2020
##  Max.   :2022
```

Key descriptive patterns from the cleaned data:

- Sustainability ratings range from 2.0 to 4.0, with a median around 3.0–3.5, indicating that most countries have relatively weak to moderate environmental policy and institutional performance.
- Gini index values span from 26.0 (low inequality) to 51.5 (high inequality), with a median around 36–37, showing meaningful variation in income inequality across countries.
- GDP per capita ranges from approximately $786 to over $23,000, demonstrating substantial cross-country economic heterogeneity. The log transformation compresses this spread into a more symmetric distribution (log GDP median 8.43).
- Years range from 2005 to 2022, with a median year of 2018. Because observations come from different years, the regression model will incorporate year fixed effects to account for temporal differences in sustainability ratings.

Regression model

We estimate the following regression:

$Sustainabilityi = _0 + _1Gini_i+ _2log(GDP_i)+ _{Year}+ _i $

This model assesses whether income inequality is associated with environmental sustainability ratings, controlling for economic development (GDP per capita) and country reporting year.

```
model_rq2_panel <- lm(
  Sustainability ~ Gini + log_GDP + factor(Year),
  data = rq2_panel
)

summary(model_rq2_panel)
```

```
##
## Call:
## lm(formula = Sustainability ~ Gini + log_GDP + factor(Year),
##     data = rq2_panel)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.82829 -0.25060 -0.02262  0.20282  1.01913
##
## Coefficients:
##                   Estimate Std. Error t value Pr(>|t|)
## (Intercept)       0.053154   0.915246   0.058  0.95389
## Gini              0.020581   0.009135   2.253  0.02820 *
## log_GDP           0.272127   0.090148   3.019  0.00382 **
## factor(Year)2006 -0.094089   0.512703  -0.184  0.85506
## factor(Year)2008 -0.327396   0.525656  -0.623  0.53592
## factor(Year)2009 -1.064825   0.522432  -2.038  0.04626 *
## factor(Year)2011  0.104357   0.371523   0.281  0.77983
## factor(Year)2012 -0.300291   0.375966  -0.799  0.42783
## factor(Year)2013  0.047017   0.345373   0.136  0.89220
## factor(Year)2014 -0.202954   0.330004  -0.615  0.54104
## factor(Year)2015  0.194192   0.335866   0.578  0.56546
## factor(Year)2016  0.654335   0.375522   1.742  0.08692 .
## factor(Year)2017 -0.238308   0.342796  -0.695  0.48981
## factor(Year)2018  0.246002   0.304130   0.809  0.42201
## factor(Year)2019  0.371635   0.292859   1.269  0.20969
```

```
## factor(Year)2020  0.662935   0.384845   1.723  0.09048 .
## factor(Year)2021  0.340690   0.294921   1.155  0.25292
## factor(Year)2022  0.160846   0.308502   0.521  0.60416
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4433 on 56 degrees of freedom
## Multiple R-squared:  0.4244, Adjusted R-squared:  0.2496
## F-statistic: 2.429 on 17 and 56 DF,  p-value: 0.006652
```

From the above table, we can deduce that:

Higher inequality is associated with higher sustainability ratings

Coefficient on Gini = 0.0206, p = 0.0288 This means that a one-point increase in the Gini index is associated with a 0.02-point increase in the sustainability rating, after controlling for GDP and year. Although the effect size is small, it is statistically significant at the 5% level.

Wealthier countries have significantly stronger sustainability ratings

Coefficient on log_GDP = 0.272, p = 0.0038 This means that a 1-unit increase in log(GDP), roughly a 2.7× increase in actual GDP per capita, is associated with a 0.27-point increase in sustainability rating. This is a moderately large substantive effect, and the p-value indicates strong significance.

Year effects are mostly small and not significant

Most year dummy coefficients lie between −0.3 and +0.3 and are far from statistical significance ($p > 0.20$), except for 2009: ***Year 2009 coefficient = −1.06, p = 0.046*** This reflects that 2009 is an unusually low outlier year which is likely due to a smaller set of countries reporting that year. Overall, year adjustment does not meaningfully change the Gini or GDP estimates, which indicates robustness.

Model fit is modest but meaningful

R^2 = 0.424, Adjusted R^2 = 0.249 This means that The model explains 42% of variation in sustainability ratings, but only 25% after adjusting for the number of predictors, though this is expected for cross-country institutional scores, which are noisy and influenced by many unmeasured factors.

To conclude, after controlling for GDP and year, income inequality shows a small but statistically significant positive association with sustainability ratings. This suggests that more unequal countries in the dataset tend to report slightly higher sustainability performance. While this pattern does not necessarily imply causation, it indicates that inequality does not depress institutional environmental ratings in the way one might expect.

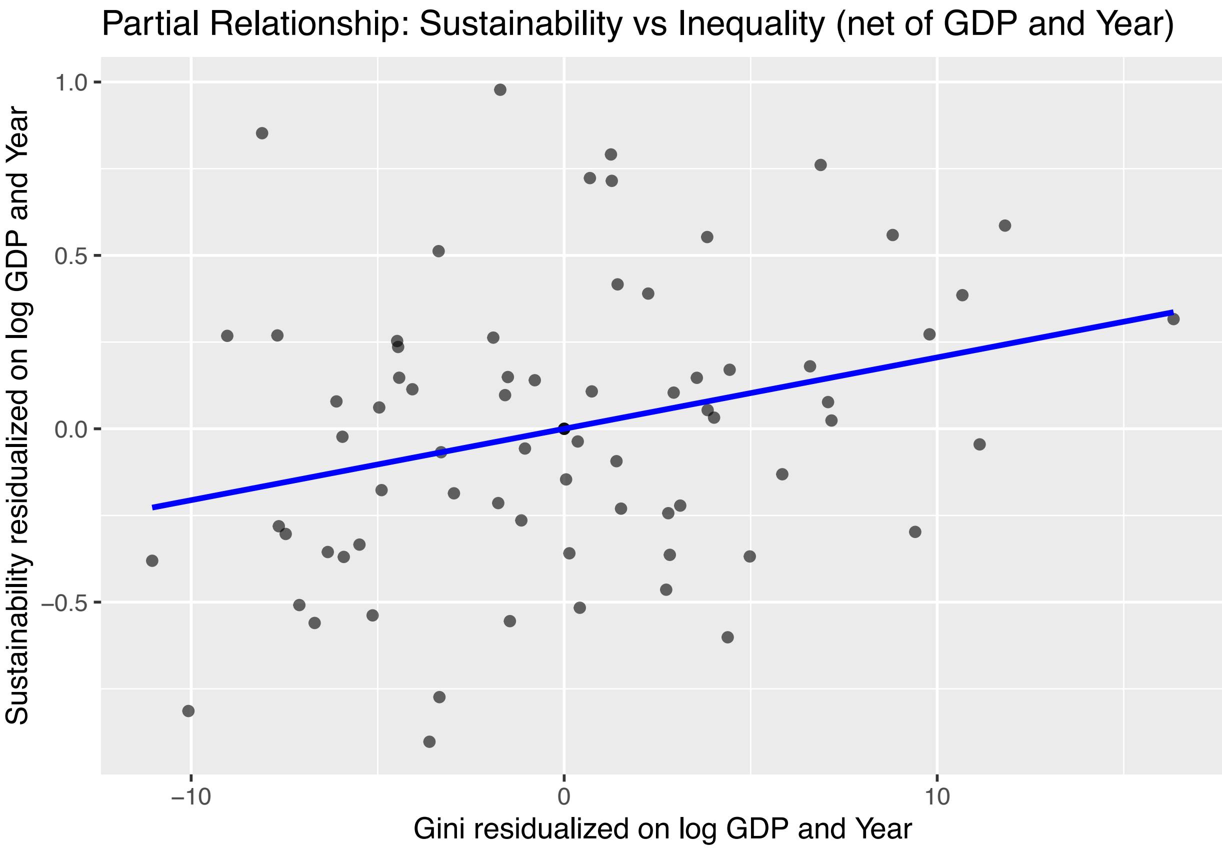
Economic development, however, displays a stronger and more consistent relationship with sustainability: countries with higher GDP per capita have substantially higher CPIA sustainability ratings. This aligns with broader empirical evidence linking economic capacity to institutional strength.

Year effects are generally not significant, suggesting that differences across reporting years do not drive the observed relationships between inequality, GDP, and sustainability.

Partial Residual Plot

To visualize the association between inequality and sustainability net of confounding factors, we constructed a partial-residual plot. Both Sustainability and Gini were residualized on log(GDP) and reporting year, and the resulting adjusted variables were plotted against each other.

```
## `geom_smooth()` using formula = 'y ~ x'
```



Key findings from the plot:

- The scatter is wide and fairly diffuse, indicating substantial variation in sustainability ratings that is not explained by inequality alone.
- Despite the noise, the fitted regression line slopes upward, echoing the positive and statistically significant coefficient from the regression model.
- Countries with higher inequality tend to have slightly higher residualized sustainability ratings even after controlling for GDP and year.
- The relationship is weak in magnitude but consistent in direction with the estimated coefficient on Gini from the main model.

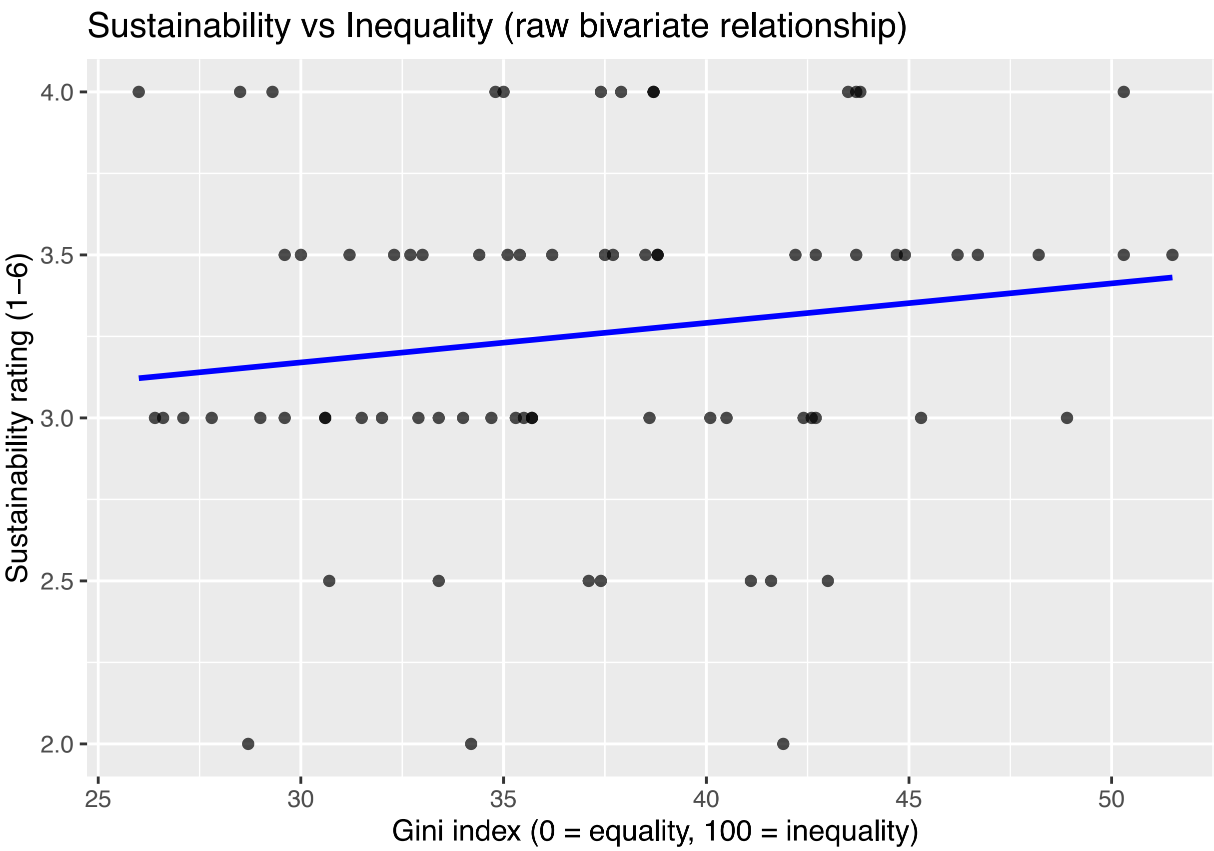
These mean that The partial-residual plot reinforces the regression finding that income inequality is positively associated with sustainability ratings, conditional on GDP and reporting year. While the association is small and the scatter shows considerable heterogeneity across countries, the upward slope indicates that inequality does not depress sustainability scores in this dataset.

However, these results do not imply causation; instead, they show that when economic development and temporal differences are accounted for, countries with higher Gini indices tend to report slightly higher environmental policy and institutional performance.

Sustainability vs Gini

```r
ggplot(rq2_panel, aes(x = Gini, y = Sustainability)) +
  geom_point(alpha = 0.7) +
  geom_smooth(method = "lm", se = FALSE, color = "blue") +
  labs(
    x = "Gini index (0 = equality, 100 = inequality)",
    y = "Sustainability rating (1-6)",
    title = "Sustainability vs Inequality (raw bivariate relationship)"
  )
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



Key findings from the plot:

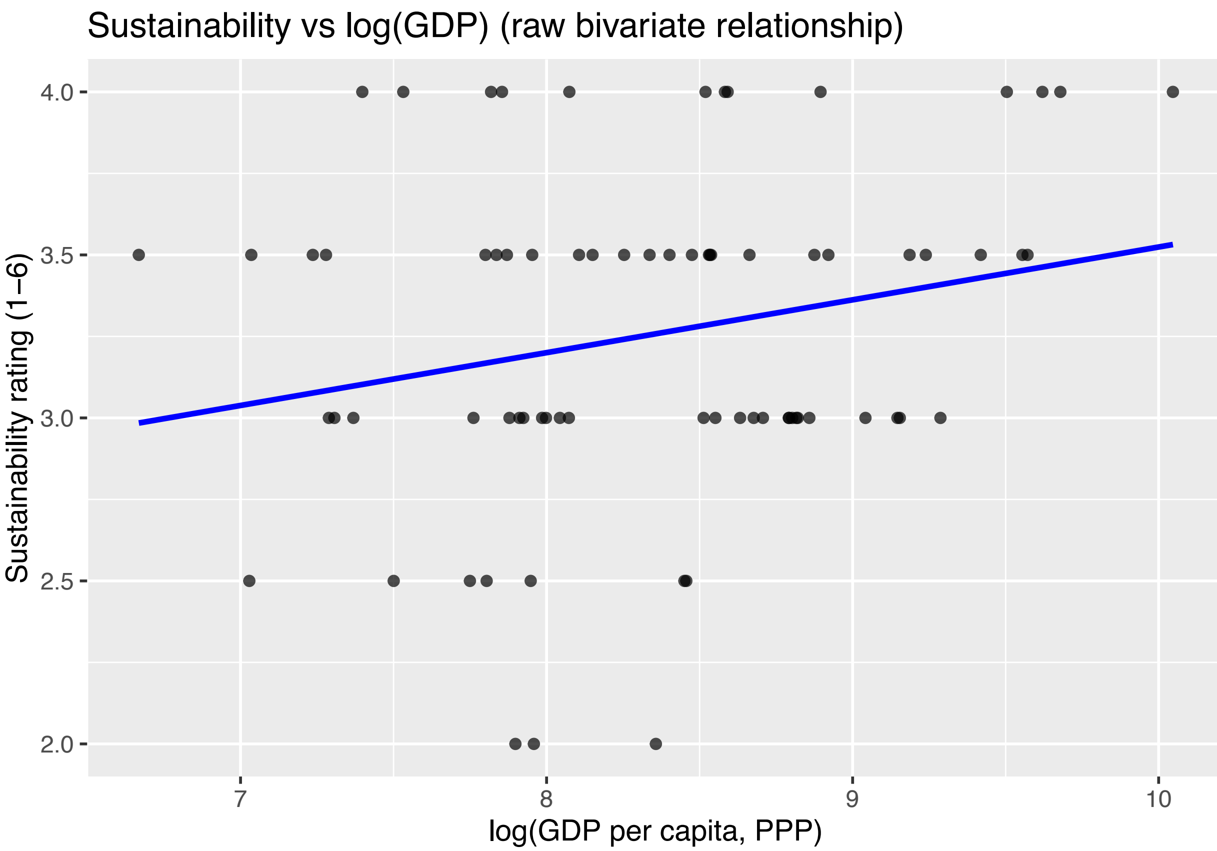
- The upward slope indicates that higher inequality is associated with slightly higher sustainability ratings in the raw data.
- The overall spread of points is wide and noisy, suggesting that inequality alone explains very little of the variation in sustainability performance.
- A large number of countries cluster at sustainability levels 3 or 3.5 regardless of inequality, indicating institutional ratings are fairly coarse.

The bivariate association mirrors the sign of the regression coefficient: more unequal countries tend to report marginally higher sustainability scores. However, the weakness of the slope and the scatter in the plot show that this relationship is not strong when examined without controls. This justifies the use of the multivariate model to account for economic development and reporting year.

Sustainability vs log(GDP)

```r
ggplot(rq2_panel, aes(x = log_GDP, y = Sustainability)) +
  geom_point(alpha = 0.7) +
  geom_smooth(method = "lm", se = FALSE, color = "blue") +
  labs(
    x = "log(GDP per capita, PPP)",
    y = "Sustainability rating (1-6)",
    title = "Sustainability vs log(GDP) (raw bivariate relationship)"
  )
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



Key findings from the plot:

- The trend line shows a stronger and more consistent positive slope than the Gini plot.
- Countries with higher log(GDP) tend to cluster at sustainability ratings of 3.5–4, while lower-income countries typically score 3 or below.
- The relationship appears more stable and less noisy than the inequality plot.

This bivariate pattern aligns closely with the regression result: wealthier countries tend to have stronger environmental policy and institutional performance. Economic capacity appears to play a more substantial role in shaping sustainability ratings than inequality does when each predictor is examined on its own.

Single year regression

Similar to RQ1, we wanted to fit a single year regression model, expecting to see improvement in data prediction. For RQ2, this corresponds to 2019, with 12 countries reporting sustainability ratings.

```r
year_counts <- rq2_panel %>%
  count(Year, name = "n_countries") %>%
  arrange(desc(n_countries))

year_counts
```

```
## # A tibble: 16 x 2
##     Year n_countries
##    <dbl>       <int>
##  1  2019          12
##  2  2021          11
##  3  2018           8
##  4  2022           7
##  5  2014           5
##  6  2015           5
##  7  2013           4
##  8  2017           4
##  9  2005           3
## 10  2011           3
## 11  2012           3
## 12  2016           3
## 13  2020           3
## 14  2006           1
## 15  2008           1
## 16  2009           1
```

The model estimated is:

\$Sustainability_{i,2019} = _0 + _1 Gini_i + _2log(GDP_i)+ _i. \$

```r
# Filter to the chosen year (2019)
rq2_2019 <- rq2_panel %>%
  filter(Year == 2019)

# Fit the cross-sectional regression
model_rq2_cross <- lm(
  Sustainability ~ Gini + log_GDP,
  data = rq2_2019
)

summary(model_rq2_cross)
```

```
##
## Call:
## lm(formula = Sustainability ~ Gini + log_GDP, data = rq2_2019)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.51785 -0.23134  0.04911  0.25453  0.47410
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
```

```
## (Intercept)  1.01484    1.51845   0.668    0.521
## Gini         0.01734    0.01435   1.208    0.258
## log_GDP      0.21678    0.13876   1.562    0.153
##
## Residual standard error: 0.3582 on 9 degrees of freedom
## Multiple R-squared:  0.2302, Adjusted R-squared:  0.05912
## F-statistic: 1.346 on 2 and 9 DF,  p-value: 0.3081
```

From the table we can see that:

- Neither Gini nor log(GDP) is statistically significant in this single-year sample.
  - Gini: estimate = 0.017, p = 0.258
  - log(GDP): estimate = 0.217, p = 0.153
- Both coefficients remain positive, matching their signs in Model 1.
- The effect sizes are similar to the panel model, but standard errors are much larger due to the smaller sample.
- The overall fit is low:
  - $R^2 = 0.23$,
  - Adjusted R^2 0.06, meaning the model explains little variation in sustainability ratings for this limited cross section.

The 2019-only regression suggests that while the direction of the relationships is consistent, showcasing by the higher inequality and higher GDP associated with slightly higher sustainability scores, the estimates are too imprecise to draw strong conclusions from a single year. The limited sample of 12 countries reduces statistical power, causing coefficients that were significant in the panel model to become non-significant here.

Therefore, Model 2 does not overturn Model 1. Instead, it shows that with only one year of data, the relationships become statistically weak even though the qualitative patterns remain the same. This reinforces the importance of using the full panel structure for institutional indicators like sustainability.

Conclusion

Across countries, sustainability ratings tend to be higher in wealthier nations and slightly higher in more unequal ones. The multivariate panel model shows that both income inequality and economic development are positively associated with environmental sustainability ratings after accounting for year differences. The effect of inequality is statistically significant but modest in size, whereas GDP demonstrates a stronger and more robust association.

The partial residual analysis indicates that, net of GDP and year, the positive relationship between inequality and sustainability persists, though with considerable variability across countries. Simple bivariate plots confirm that GDP is a more substantial predictor of sustainability than inequality when each is examined independently.

The single-year model for 2019, although directionally consistent with the panel results, suffers from limited sample size and produces imprecise estimates. As a result, the evidence supporting the role of inequality is clearer in the full dataset than in a single-year cross-section. Overall, economic development appears to be the key driver of sustainability ratings, while inequality shows a small but positive association that warrants cautious interpretation.